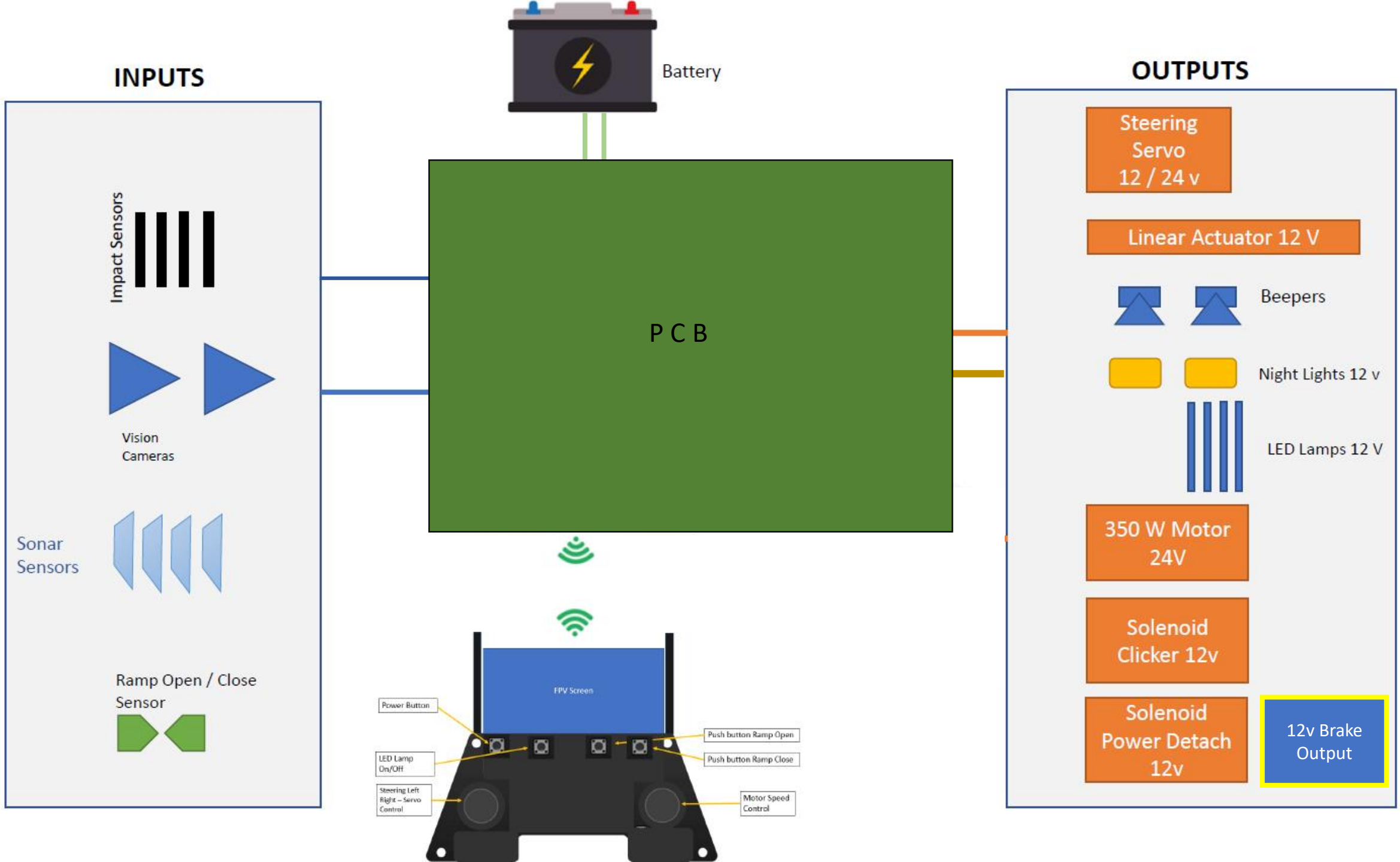
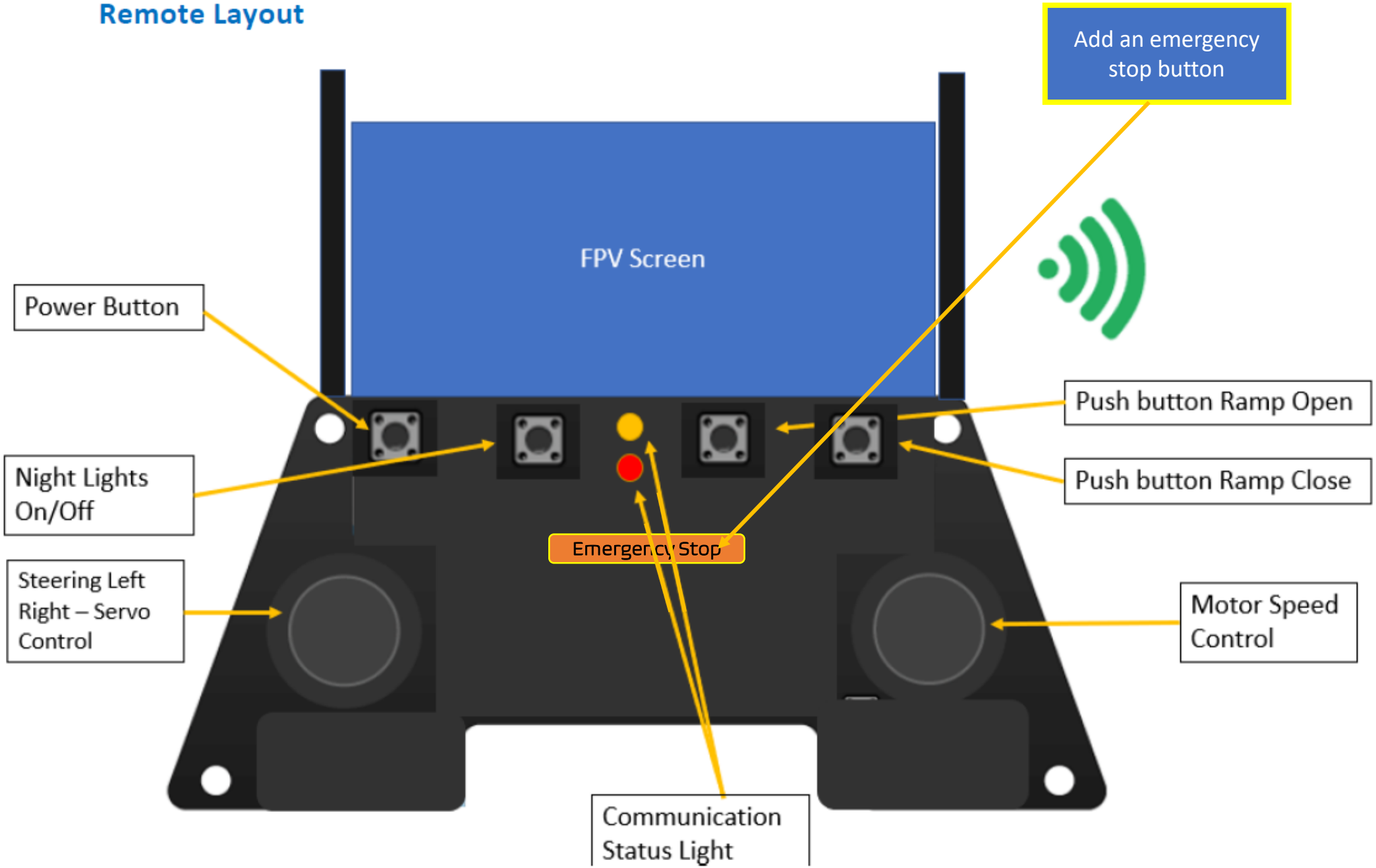
Trobo –Lite

Firmware Code Requirements

PCB Connectors


INPUTS
Battery
OUTPUTS
Impact Sensors
P C B
Steering Servo 12 / 24 v
Linear Actuator 12 V
Beepers
Night Lights 12 v
LED Lamps 12 V
Vision Cameras
Sonar Sensors
350 W Motor 24V
Solenoid Clicker 12v
Ramp Open / Close Sensor
Solenoid Power Detach 12v
12v Brake Output
FPV Screen
Power Button
Push button Ramp Open
Push button Ramp Close
LED Lamp On/Off
Steering Left Right – Servo Control
Motor Speed Control

Remote Layout



PCB Requirements

- Provide precise and smooth control of permanent magnet drive motors
  - 24v Transaxle
  - Linear actuator
  - Servo Motor

- Allow adjustment to configurations for output (outlined in the configuration criteria)
- Charger inhibit – prevent maneuvering while charger is attached
- Internal main relay provides secure power-off
- Power Saver function to prevent battery drain when the cart is inactive
- Controller power circuits and microprocessor software should be continuously monitored for proper operation
- Loss of RF signal should stop all cart operations tills signal has been reestablished for at least 5 seconds
- Power-up, auto system checks to ensure that associated wiring is intact, and disables drive if a fault is found
- Output Errors on an embedded LCD Shield Kit 16x2 Character Display screen.

24v Motor Operation

- Maintains speed over varied terrain, obstacles, curbs, and ramps
- Linear cutback of there should not be sudden loss of power during undervoltage or overtemperature conditions
- Anti-Rollback/Roll-forward function to provides smooth and safe vehicle control on hills and ramps - let me know if we need pot input for this feature
- Boost current - a brief boost of current for starting on a hill, crossing thresholds, etc - let me know if we need pot input for this feature

PCB Adjustments

- PCB adjustments for input and output can be adjusted using following three options:
  - RS232 Interface: This will allow to digitally input values in a gui to adjust the parameters outlined on next page.
  - Providing onboard jumpers and configurations that allow for adjustment
  - Providing onboard pots that can be adjusted manually – old analog approach.
  - Other option that I am not aware of – anything that you recommend – Bluetooth, wifi, ssh?

Adjustable Parameters on PCB via RS232

DRIVE MOTOR

RAMP MOTOR

SERVO MOTOR

SOLENOID

LIGHTS

BEEPER

CAMERA

RF

BRAKE

SONAR

Adjustable Parameters on PCB

DRIVE MOTOR

Parameter	Allowable Range	Description
Accel Speed (Forward/Reverse)	0.5 – 5.0 s	Sets the rate (in seconds) at which the speed command increases when throttle is applied from remote with the speed limit throttle is in its maximum speed position. Larger values represent slower response
Decel Speed (Forward/Reverse)	0.5 – 5.0 s	Sets the rate (in seconds) that is used to slow down the cart when it is traveling forward at high speed and throttle is reduced. Larger value represent slower response.
E-stop	0.5 -2 s	Sets the rate (in seconds) that is used to slow down the vehicle during emergency or autonomous stop scenario (Sonar/Clash sensor)
Soft Start	0–100 %	This parameter can be used to soften the bump associated with gear slack in the transaxle when throttle is applied, released and then reapplied from the neutral state. Larger values provide a softer slack take-up.

Adjustable Parameters on PCB

RAMP MOTOR

Parameter	Allowable Range	Description
Max Speed	0–100 %	During forward/ reverse operation, defines the requested speed for opening and closing.
Max Actuator Open	0–100 %	Maximum opening length of the actuator
Min Actuator Close	0–100 %	Length at which the actuator should stop during retraction / closure
Open Contact Senor	On / Off	The actuator is disengaged during operation when the open contact sensor is engaged. Off disengages the contact sensor feedback and allow the actuator to open at max actuator open speed.
Close Contact Senor	On/ Off	The actuator is disengaged during close operation when the open contact sensor is engaged. Off disengages the contact sensor feedback and allow the actuator to open at max actuator open speed.
Stall Speed	0-100	Sets the Stall Speed that will be used to determine a Motor Stalled fault.
Stall Timeout	0.5- 5 S	Sets the Stall Timeout that will be used to determine a Motor Stalled fault;

Adjustable Parameters on PCB

SERVO MOTOR

Parameter	Allowable Range	Description
Max Left Angle	0–100	Each value defines a 1 degree angle allowing up to 100 degree on the left.
Max Right Angle	0–100	Each value defines a 1 degree angle allowing up to 100 degree on the right.
Contact Sensor	On/Off	Contact Sensor on left and right defines the maximum limit of the servo rotation. Servo cannot rotate further than the contact sensor. Contact Sensor On – disengages or overrules the max left and right angle.
Stall Speed	0-100	Sets the Stall Speed that will be used to determine a Servo Stalled fault.
Stall Timeout	0.2- 5 S	Sets the Stall Timeout that will be used to determine a Servo Stalled fault.

See Servo Spec Sheet for additional signaling input and jumper configurations.
File Name: 34109MDData.pdf

Adjustable Parameters on PCB

SOLENOID

Parameter	Allowable Range	Description
Operating Pulse	0.5 – 5 S	Operating Pulse defines the pulse for which the solenoid will get current on command from the remote.
Stall Timeout	0.2- 5 S	Sets the Stall Timeout that will be used to determine a Solenoid Stuck fault.

Adjustable Parameters on PCB

LIGHTS

Parameter	Allowable Range	Description
Night Lights*	0.5 – 10 S	Each value defines a the second for which the lights will remain on once the cart has stopped. Night Lights should have a soft on and soft off.
LED Lamps	0.5 – 10 S	Each value defines a the second for which the lights will remain on once the cart has stopped.
LED Operation	On/Off/Blinking	Defines whether LED Lights will work as On/ Off or Blinking with 1 second on and 1 second off.

Adjustable Parameters on PCB

BEEPER

Parameter	Allowable Range	Description
Operation	On/Off	Turn on Beeper / Deactivate Beeper
Beeper Voltage	1 - 3	1 is 6v 2 is 9v 3 is 12v
Beeper Operation	On/Off/Blinking	Defines whether LED Lights will work as On/ Off or Blinking with 1 second on and 1 second off.

Adjustable Parameters on PCB

CAMERA

Parameter	Allowable Range	Description
Operation	On/Off	Turn on the camera on or off during operation.
Reverse	Yes or No	Yes, is forward / reverse camera reversed with operations. Forward drive operates rear camera. Rear Drive operates forward camera.
Camera Delay	0.5 - 10 s	Defines the time limit for which the camera should after the cart has been stopped.

Adjustable Parameters on PCB

RF

Parameter	Allowable Range	Description
Channel Selection	1 …n	Select Applicable channels for RF communication
Program /Pair New Remote	Yes or No	Allows user to pair new remote controller.

Adjustable Parameters on PCB

Brake

Parameter	Allowable Range	Description
Hold Voltage	10–24 V	A high initial voltage is applied to the brake coil when the brake is first released. After approximately 1 second, this peak voltage drops to the programmed Hold Voltage. The parameter should be set high enough to hold the brake released under all the shock and vibration conditions the Cart will be subjected to.
Fault Check	On / Off	Enables/disables the fault detection on the EM brake.
Delay	0–2 s	Sets the length of delay between when zero speed is commanded and the electromagnetic brake is engaged.